APEX RETAIL GROUP | PRACTICAL CLASS TASKS

DOCUMENT: CLASS TASKS (SESSION 8)

SCENARIO: CHECKPOINT 1 — Q3 PROFITABILITY & OPERATIONAL LEAKAGE INVESTIGATION

INSTRUCTIONS FOR LEARNER: This is your Mid-Course Checkpoint. You must execute a complete, end-to-end Exploratory Data Analysis (EDA) on the provided dataset without step-by-step code guidance. Follow the structured 5-stage EDA workflow: **Data Audit -> Sanitization -> Relational Merge -> Investigation -> Business Briefing.**

DATASETS PROVIDED

You are provided two raw data dictionaries representing Q3 operations:

```Python
import pandas as pd
import numpy as np

# File 1: Raw Q3 Transactions
raw_q3_txns = {
    "txn_id": ["TX-101", "TX-102", "TX-103", "TX-103", "TX-104", "TX-105", "TX-106", "TX-107", "TX-108"],
    "facility_id": ["FAC-A", "FAC-B", "FAC-C", "FAC-C", "FAC-A", "FAC-B", "FAC-C", "FAC-A", "FAC-B"],
    "txn_date": ["2026-07-01", "2026-07-03", "2026-07-10", "2026-07-10", "2026-08-05", "2026-08-12", "2026-09-01", "2026-09-15", "2026-09-28"],
    "product_line": [" FOOTWEAR ", "Apparel", " accessories ", " accessories ", "Footwear", " APPAREL ", "Accessories", "Footwear", "Apparel"],
    "gross_sales": [1200.00, 350.00, np.nan, np.nan, 850.00, 420.00, 190.00, 950.00, 310.00],
    "units_shipped": [10, 5, 2, 2, 8, 6, 3, 9, 4]
}

# File 2: Facility Operating Metadata
raw_facility_meta = {
    "facility_id": ["FAC-A", "FAC-B", "FAC-C"],
    "region_name": ["Northeast", "Southeast", "Midwest"],
    "fulfillment_cost_per_txn": [250.00, 180.00, 220.00]
}
```

TASK 1: DATA AUDIT & SANITIZATION PIPELINE

1. Load both dictionaries into Pandas DataFrames (`df_txns` and `df_facilities`).
2. Audit `df_txns`: Identify missing values and exact duplicate rows.
3. Clean `df_txns`:
    - Remove duplicate rows and reset index.
    - Standardize `product_line` (strip whitespace and convert to Title Case).
    - Convert `txn_date` to `datetime64[ns]`.
    - Impute missing values in `gross_sales` using the median value of valid entries.

TASK 2: RELATIONAL MERGE & FEATURE ENGINEERING

1. Merge `df_txns` (left) with `df_facilities` (right) on `"facility_id"` using a Left Join.
2. Create a new calculated feature column: `net_margin = gross_sales - fulfillment_cost_per_txn`.
3. Print the merged DataFrame and confirm no `NaN` values exist.

TASK 3: MULTI-DIMENSIONAL EDA INVESTIGATION

1. Group the merged dataset by `region_name` and calculate:
    - Total gross sales (`gross_sales`: `sum`)
    - Total fulfillment costs (`fulfillment_cost_per_txn`: `sum`)
    - Total net margin (`net_margin`: `sum`)
    - Order count (`txn_id`: `count`)
2. Sort the regional summary table by `net_margin` in ascending order to highlight underperforming locations.
3. Group by `product_line` and calculate total `gross_sales` and average `net_margin`.

TASK 4: EXECUTIVE ADVISORY BRIEFING

Review your aggregated investigation outputs from Task 3:

1. What is the total audited gross sales figure across all facilities?
2. Which specific region represents the primary source of profit leakage, and why?
3. Draft a 150-word formal executive briefing memo to Victoria Vance (VP of Operations) summarizing your EDA findings and offering two evidence-based strategic recommendations.